



The real effects of AI: Evidence from corporate investment efficiency[☆]

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ABSTRACT

This study examines how artificial intelligence (AI) adoption is associated with corporate investment efficiency among U.S. firms. We measure AI adoption using *AI mentions*, defined as the proportion of AI-related terminology in senior management remarks during earnings calls, and validate this measure by showing that it is positively associated with 10-K discussions of AI applications in operational decision-making, product development, and business expansion. We construct an instrumental variable based on exposure to university-authored AI research—measured as a normalized score of textual similarity between non-corporate academic publications and industry descriptions—to address concerns that AI adoption may be endogenous to firms' investment decisions and investment efficiency. Our baseline results indicate that higher AI adoption is associated with greater investment efficiency, and this association remains in two-stage least squares regressions using the AI technology exposure instrument. We also find that firms with higher AI mentions tend to have higher Tobin's *q* over longer horizons, consistent with a gradual reflection of AI-related benefits in firm value. Additional analyses suggest several mechanisms: AI adoption is associated with more accurate management sales forecasts, higher financial reporting quality, and greater process and product innovation intensity. Cross-sectional evidence further indicates that regulatory frictions, such as exposure to data-privacy regulation, attenuate the association between AI adoption and investment efficiency. Overall, the findings suggest that AI adoption is an emerging factor in corporate investment and capital allocation.

1. Introduction

Recent advances in artificial intelligence (AI) are changing how firms process information for real investment decisions. Tools based on machine learning and natural language processing can analyze large, complex datasets and, in some settings, generate risk assessments and predictions that systematically outperform human judgment given the same information (Kleinberg et al., 2018;

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